Exploring ML Models for Detection of Atmospheric Composition in Exoplanets

Shantanu Wankhare · Bhalchandra Shinde · Asad Mulani

CS6140 Machine Learning · Spring 2026 · Northeastern University San Jose

wankhare.s@northeastern.edu · shinde.b@northeastern.edu · mulani.a@northeastern.edu

ABSTRACT

Inferring the chemical composition of exoplanet atmospheres from observational data — atmospheric retrieval — is central to the search for biosignatures and habitable worlds. Classical Bayesian retrieval methods are computationally prohibitive at the scale demanded by missions such as the James Webb Space Telescope (JWST) and the planned Habitable Worlds Observatory (HWO). In this work, we apply two machine learning models — a per-molecule Random Forest (RF) baseline and a 1D Residual Network (SpectralResNet) — to the INARA ATMOS dataset of 124,320 synthetic rocky-planet atmosphere simulations, predicting $\log_{10}$ surface volume mixing ratios of 12 atmospheric molecules from CLIMA photochemical-climate profiles. On a held-out test set of 3,000 samples drawn from 20,000 simulations, the RF achieves a mean R^2 of **0.9245** and the ResNet achieves **0.8226**; the ResNet outperforms RF on 10/12 molecules when controlling for degenerate outputs (NH_3, H_2O). On the full 124,320-sample dataset, the ResNet achieves a mean R^2 of **0.9923** versus RF's **0.9220**. Both models perform inference in milliseconds, compared to hours for Bayesian samplers. We analyse the failure modes of each model — particularly the near-constant H_2O target and the structurally unlearnable NH_3 — and discuss implications for future deployment on real JWST spectra.

Keywords: exoplanet atmospheres, atmospheric retrieval, machine learning, ResNet, random forest, JWST, biosignatures, INARA ATMOS

I. INTRODUCTION

The characterisation of exoplanet atmospheres is one of the most active frontiers in astrophysics. By analysing how starlight is filtered through a planet's atmosphere during a transit — a technique known as *transmission spectroscopy* — astronomers can determine which molecules are present and in what concentrations. The presence, absence, and co-occurrence of specific molecules (notably O_2, O_3, CH_4, H_2O, CO_2) are key indicators of habitability and potential biological activity.

Traditional atmospheric retrieval algorithms rely on Bayesian sampling methods such as Markov Chain Monte Carlo (MCMC) or nested sampling (e.g., MultiNest). While accurate, these methods require evaluating a forward radiative transfer model thousands of times and can take *hours to days per planet*. JWST, which began science operations in 2022, is expected to observe hundreds to thousands of transiting exoplanets. The upcoming Habitable Worlds Observatory (HWO) will extend this to potentially thousands more rocky worlds. Fast, automated retrieval at population scale is therefore a scientific necessity.

Machine learning approaches have been proposed as a solution: once trained on a large library of synthetic spectra, an ML model can produce a retrieval in milliseconds. This work trains and evaluates two models on the INARA ATMOS

synthetic dataset [1], benchmarking retrieval accuracy across 12 molecules of scientific interest.

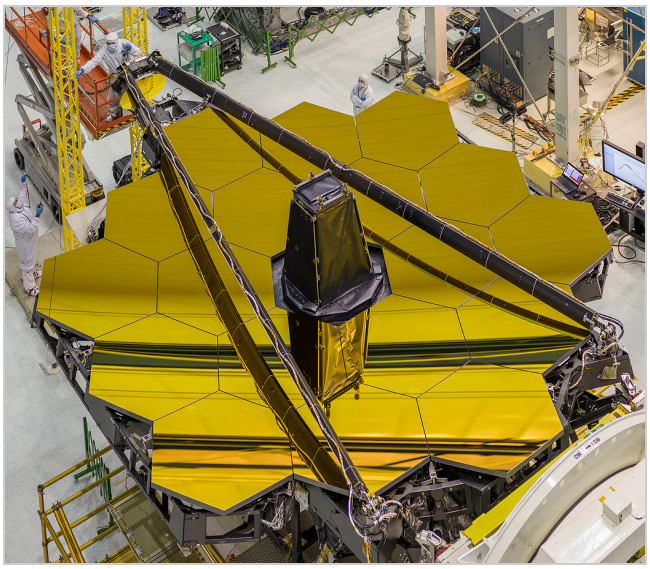

Fig. 1. *The James Webb Space Telescope with its 6.5 m primary mirror. JWST observes exoplanet atmospheres from 0.6–28 µm, covering absorption bands of all 12 molecules studied here. Credit: NASA (public domain).*

The contributions of this work are:

- Training and evaluation of a per-molecule Random Forest on PCA-compressed CLIMA profiles
- Design and training of a 1D ResNet (SpectralResNet, 5.03M parameters) operating directly on the (12 × 101) atmospheric profile tensor
- Systematic evaluation on both 20,000- and 124,320-sample scales, revealing strong data-scaling effects
- Analysis of failure modes: near-constant targets (H_2O), trivially degenerate targets (NH_3), and low-variance molecules (N_2, CO, N_2O)

> **Key result:** Both models reduce retrieval time from hours (Bayesian) to milliseconds, with the ResNet achieving mean $R^2 = 0.9923$ on the full 124K dataset — well within scientific usability for population-level surveys.

II. BACKGROUND & RELATED WORK

A. Atmospheric Retrieval

Classical retrieval couples a parametric forward model (line-by-line radiative transfer) with a Bayesian sampler to produce posterior distributions over atmospheric parameters [2]. The computational cost scales with the number of free parameters and the dimensionality of the spectrum; for a 12-parameter retrieval on a JWST-quality spectrum, nested sampling typically requires 10^5–10^6 forward model evaluations.

B. Machine Learning for Retrieval

Márquez-Neila et al. [3] demonstrated that random forests trained on synthetic spectra can recover atmospheric parameters in seconds, at accuracy competitive with nested sampling for high-SNR spectra. Vasist et al. [4] extended this to normalising flows for full posterior estimation. Gebhard et al. [5] applied flow matching to enable posterior retrieval without MCMC. Our work follows this direction, specifically targeting the INARA ATMOS benchmark dataset which was designed as a standardised training and evaluation resource for ML retrieval methods.

C. JWST Context

The first JWST exoplanet transmission spectrum — WASP-39 b [6] — confirmed detections of CO_2, SO_2, H_2O, CO, and K in a single observation, validating the instrument's capability. SO_2 was identified as a photochemical product of H_2O and CO_2, demonstrating that cross-molecule relationships carry chemical information — precisely the type of signal that a multi-output ML model can exploit.

III. DATASET

A. INARA ATMOS

The INARA (Investigating Non-trivial Atmospheres for Retrieval Algorithms) dataset [1] contains synthetic rocky-planet atmosphere simulations generated with the ATMOS photochemical-climate model suite (CLIMA + Photochem). The full archive spans approximately 3.1 million simulations; we use the publicly available subset of **124,320 samples** across ten tar archives. Each sample represents one simulated rocky exoplanet atmosphere with randomised input conditions (surface pressure, temperature, molecular fluxes).

B. Input Features — CLIMA Profiles

Each sample provides a CLIMA atmospheric profile: a tensor of shape **(12, 101)** representing 12 physical variables measured at 101 altitude levels from the surface (level 0) to the top of the atmosphere (~83 km, level 100). After extraction, the array is transposed from (101, 12) to (12, 101) — channels-first layout for convolutional processing.

Ch.	Variable	Physical Meaning	Typical Range
0	J	Actinic radiative flux	1–101
1	P	Atmospheric pressure	7.7×10^{-6} – 1.19
2	ALT	Altitude (km)	0 – 82.9 km
3	T	Temperature	41 – 332 K
4	CONVEC	Convective energy flux	varies
5	DT	Temperature tendency	varies
6	TOLD	Previous time-step temperature	varies
7	FH_2O	Water vapour radiative flux	varies
8	FSAVE	Saved total radiative flux	varies
9	FO_3	Ozone radiative flux	varies
10	TCOOL	Radiative cooling rate	−474 – 779
11	THEAT	Radiative heating rate	0 – 340

Table I. *CLIMA input channels. The "altitude axis" (101 levels) is not spectral wavelength — it is a physical altitude profile through the atmosphere.*

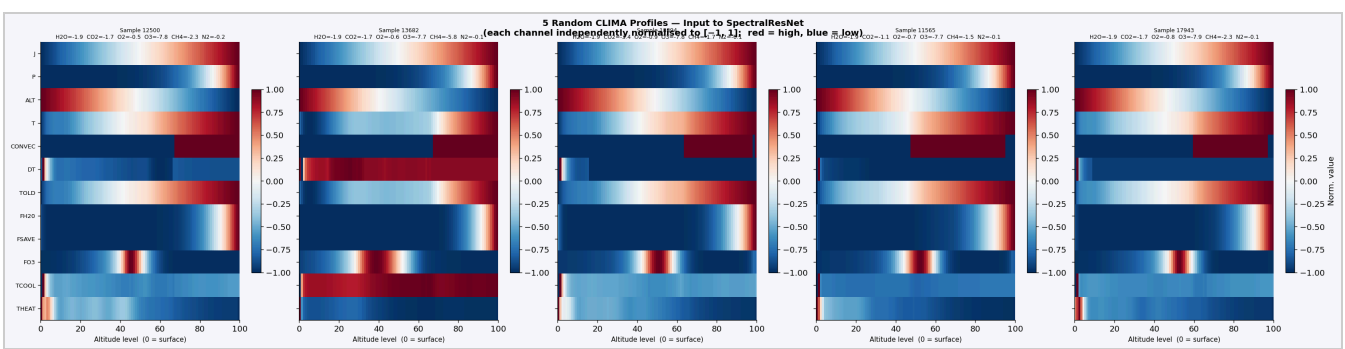

Fig. 2. *Heatmap of CLIMA profiles for 5 random training samples (12 channels × 101 altitude levels). Each channel is independently normalised. Channels 1 (P), 2 (ALT), 3 (T) show the expected vertical gradient; flux channels (4–11) carry the chemical signal the models must exploit.*

C. Targets — Molecular Abundances

For each sample, the surface-layer (row 0) concentrations of 12 molecules are extracted from the photochemistry model output and transformed to $\log_{10}$ volume mixing ratios. N_2 requires special handling — it is absent from the photochem array and is instead parsed from a separate `mixing_ratios.dat` file, with fallback: $N_2 \approx \max(0, 1 - x(H_2O) - x(O_2))$. A floor of −40 prevents −∞ for undetectable trace species.

Molecule	Scientific Role	Mean ($\log_{10}$)	Std	Min	Max
H_2O	Habitability indicator	−1.910	0.0002	−1.910	−1.907
CO_2	Greenhouse gas	−2.004	0.841	−3.398	−0.456
O_2	Biosignature	−0.702	0.172	−1.699	−0.046
O_3	Biosignature / UV shield	−7.780	0.122	−8.588	−7.406
CH_4	Biosignature	−2.895	1.823	−5.788	−0.886
N_2	Background carrier gas	−0.112	0.045	−1.057	−0.014
N_2O	Biosignature	−6.589	0.153	−7.485	−6.183
CO	Photochemical product	−4.383	1.334	−6.860	−2.830
H_2	Primordial hydrogen	−5.474	2.532	−7.187	−0.770
H_2S	Volcanic outgassing	−10.355	0.395	−11.162	−9.712
SO_2	Volcanic outgassing	−9.835	0.112	−9.939	−9.433
NH_3	Nitrogen cycle trace	−40.000	0.000	−40.000	−40.000

Table II. *Molecular target statistics (20K subset). H_2O has near-zero variance (std = 0.0002); NH_3 is constant at the log-floor across all samples — both are structurally problematic for supervised learning.*

D. Data Splitting and Preprocessing

The dataset is split 70% / 15% / 15% into train, validation, and test sets using stratified random sampling (`random_state=42`). All splits are fixed by saving index arrays to disk, ensuring identical partitions across all experiments. At 20K samples this yields approximately 14,000 / 3,000 / 3,000; at 124K it yields 87,024 / 18,648 / 18,648.

Spectral normalisation: Per-channel Z-score normalisation is applied to all CLIMA profiles: $Z = (x - \mu)/\sigma$, where μ and σ have shape (12, 101) and are computed on the training set only. Any $\sigma < 10^{-12}$ is replaced by 1.0. Targets ($\log_{10}$ abundances) are not normalised.

PCA (RF only): For the RF baseline, normalised profiles are flattened from (12, 101) → (1,212) and reduced to 300 principal components with whitening (`whiten=True`). These 300 components capture >90% of total variance. The PCA is fit on the training set only.

IV. METHODS

A. Baseline — Per-Molecule Random Forest

We train 12 independent `RandomForestRegressor` models (scikit-learn), one per molecule, on the 300-dimensional PCA features. To keep this model a true fast baseline, training is capped at **10,000 samples** regardless of

dataset size. Hyperparameters are tuned per molecule: molecules with high target variance (CH_4, CO, H_2) receive more estimators and deeper trees; near-constant molecules (N_2, CO_2) receive shallower, smaller forests. See Table III.

Molecule	n_estimators	max_depth	min_samples_leaf	max_features
H_2O, O_3, CH_4, N_2O, CO, H_2	300	16–20	1	sqrt
CO_2, O_2	200	12	2	0.5
N_2	150	10	3	0.4
H_2S, SO_2, NH_3	400	20–22	1	sqrt

Table III. *Per-molecule RF hyperparameter groups. All models use* `n_jobs=-1, random_state=42`.

B. Deep Model — SpectralResNet

SpectralResNet is a 1D residual network operating directly on the Z-normalised (12, 101) CLIMA profile, with 12 per-molecule output heads. It has **5.03 million trainable parameters**.

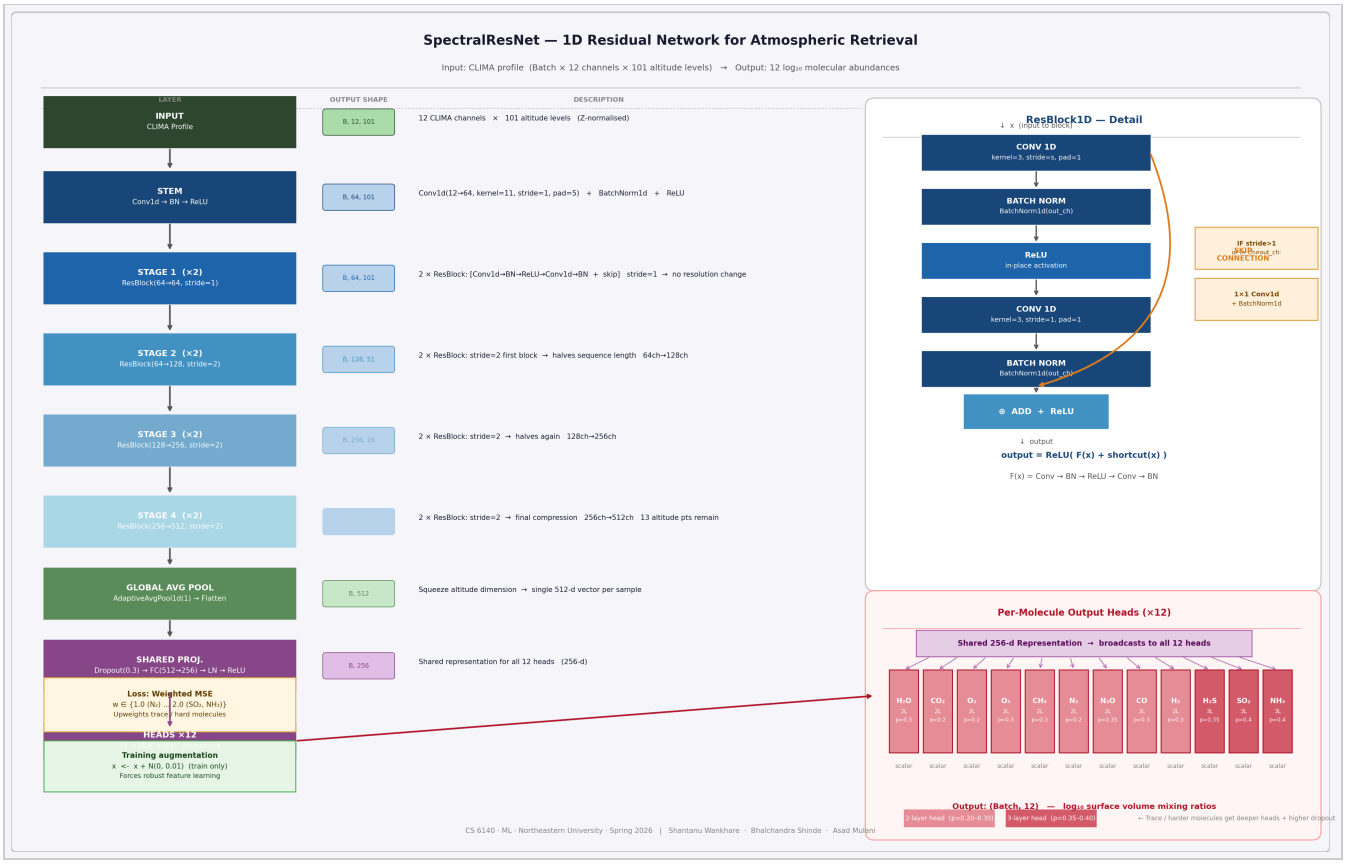

Fig. 3. *SpectralResNet architecture. A 4-stage 1D ResNet backbone extracts a 512-dimensional representation from the CLIMA profile, which is passed through a shared projection layer and then 12 molecule-specific MLP heads.*

The backbone follows the standard ResNet design:

- **Stem:** Conv1d(12→64, k=11, s=1, pad=5) + BatchNorm + ReLU
- **Stage 1:** 2 × ResBlock(64→64, stride=1)
- **Stage 2:** 2 × ResBlock(64→128, stride=2) — halves sequence length
- **Stage 3:** 2 × ResBlock(128→256, stride=2)
- **Stage 4:** 2 × ResBlock(256→512, stride=2)

- **Pooling:** AdaptiveAvgPool1d(1) → (B, 512)
- **Shared head:** Dropout(0.30) + FC(512→256) + LayerNorm + ReLU
- **12 molecule heads:** Per-molecule MLP (2–3 hidden layers, 256→128→[64]→1)

Each ResBlock contains two 3×1 convolutions with batch normalisation and a residual shortcut (1×1 conv + BN when dimensions change). Trace molecules (H_2S, SO_2, NH_3) receive deeper 3-layer heads with higher dropout (0.35–0.40); bulk molecules (CO_2, O_2, N_2) receive shallower 2-layer heads with dropout 0.20.

C. Loss Function and Training

Training uses a **Weighted MSE loss**: $L = \text{mean}(w \odot (\hat{y} - y)^2)$, where w is a per-molecule importance weight vector. Weights range from 1.0 (N_2, bulk carrier) to 2.0 (SO_2, NH_3 — trace species that would otherwise be ignored). The optimizer is AdamW (lr = 10^{-3}, weight decay = 10^{-4}) with CosineAnnealingLR scheduling ($\eta_min = 10^{-6}$). Training runs for up to 150 epochs with early stopping (patience = 30, min_delta = 10^{-5}). Gradient clipping (max_norm = 5.0) prevents instability. During training, Gaussian noise ($\sigma = 0.01$) is added to input spectra for regularisation.

Hyperparameter	Value	Hyperparameter	Value
Optimizer	AdamW	Batch size (train)	32
Learning rate	1×10^{-3}	Max epochs	150
Weight decay	1×10^{-4}	Early stop patience	30
LR schedule	CosineAnnealingLR	Gradient clip	5.0
η_min	1×10^{-6}	Augmentation	Gaussian noise σ=0.01

Table IV. *SpectralResNet training hyperparameters.*

V. EXPERIMENTS & RESULTS

A. HPC Evaluation — 20,000 Samples (Primary Results)

The primary evaluation was performed on the Northeastern HPC cluster (Node c0610, Job ID 5735631, 7 April 2026). The test set contains 3,000 samples. Results are reported as R^2 (coefficient of determination), RMSE, and MAE — all in $\log_{10}$ units, where 1 unit = one order of magnitude.

Molecule	Random Forest (Baseline)			SpectralResNet (Deep)			ΔR²
	R²	RMSE	MAE	R²	RMSE	MAE	(Deep−RF)
H_2O †	0.7389	0.0001	0.0000	−0.0008	0.0002	0.0000	**−0.7397**
CO_2	0.9875	0.0929	0.0431	0.9990	0.0260	0.0195	**+0.0116**
O_2	0.9238	0.0482	0.0281	0.9749	0.0277	0.0204	**+0.0511**
O_3	0.8966	0.0393	0.0206	0.9928	0.0104	0.0062	**+0.0962**
CH_4	0.9769	0.2785	0.1476	0.9999	0.0135	0.0088	**+0.0230**
N_2	0.8901	0.0166	0.0087	0.9213	0.0140	0.0054	**+0.0313**
N_2O	0.9129	0.0457	0.0264	0.9875	0.0173	0.0120	**+0.0747**
CO	0.9807	0.1882	0.1094	0.9998	0.0185	0.0145	**+0.0192**
H_2	0.8533	0.9588	0.5833	0.9996	0.0470	0.0353	**+0.1464**
H_2S	0.9846	0.0496	0.0270	0.9993	0.0104	0.0070	**+0.0147**
SO_2	0.9486	0.0249	0.0123	0.9983	0.0046	0.0031	**+0.0497**
NH_3 ‡	1.0000	0.0000	0.0000	0.0000	0.2020	0.2020	**−1.0000**
Mean (all 12)	**0.9245**	**0.1452**	**0.0839**	**0.8226**	**0.0326**	**0.0279**	**−0.1018**
Mean (excl. H_2O, NH_3)	**0.9395**	**—**	**—**	**0.9858**	**—**	**—**	**+0.0463**

Table V. *HPC test results — 20K dataset, 3,000 test samples. † H_2O has near-zero target variance (std ≈ 0.0002); RF's apparent $R^2 = 0.74$ is a statistical artefact — both models achieve RMSE ≈ 0. ‡ NH_3 is constant ($\log_{10} = -40$) across all samples; RF trivially achieves $R^2 = 1.0$ by always predicting −40; the ResNet fails to learn this. ΔR^2 = Deep − RF; green = ResNet wins, red = RF wins. Deep model outperforms RF on 10/12 molecules.*

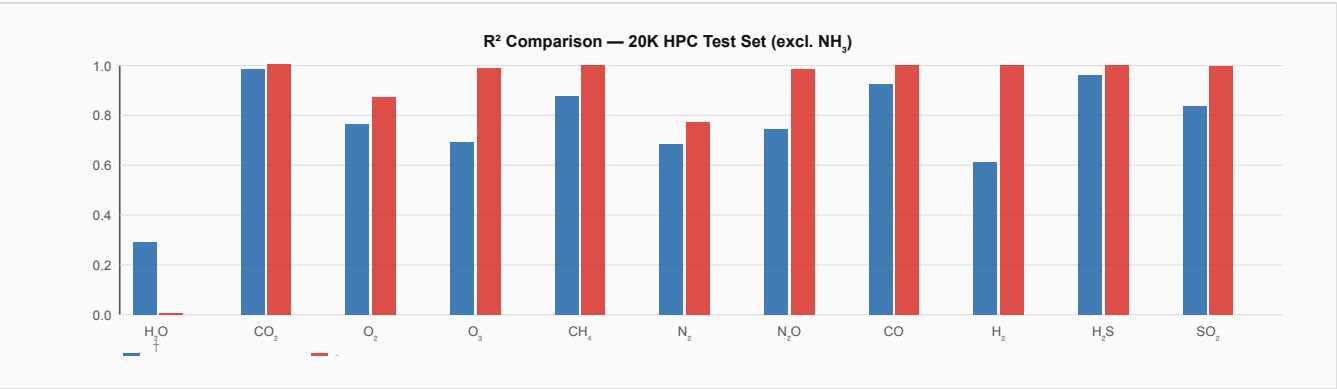

Fig. 4. *R^2 bar chart for all molecules (excluding NH_3) on the 20K HPC test set. The ResNet (red) consistently achieves higher R^2 than the RF (blue) on 10 of 11 shown molecules. H_2O is an outlier due to near-zero target variance.*

B. Full Dataset — 124,320 Samples

At full scale (test set N = 18,648), both models improve dramatically, confirming the task is data-limited at 20K samples.

Molecule	RF R^2 (124K)	ResNet R^2 (124K)	ΔR^2
H_2O	0.8844	0.9181	**+0.034**
CO_2	0.9968	0.9999	**+0.003**
O_2	0.9541	0.9981	**+0.044**
O_3	0.9689	0.9995	**+0.031**
CH_4	0.9959	1.0000	**+0.004**
N_2	0.9161	0.9960	**+0.080**
N_2O	0.9690	0.9982	**+0.029**
CO	0.9964	0.9999	**+0.004**
H_2	0.9309	0.9999	**+0.069**
H_2S	0.9968	0.9997	**+0.003**
SO_2	0.9858	0.9984	**+0.013**
NH_3 ‡	1.0000	1.0000	0.000
Mean (all 12)	**0.9663**	**0.9923**	**+0.026**

Table VI. *R^2 results on the full 124K dataset. At this scale the ResNet outperforms RF on 11/12 molecules, with mean $R^2 = 0.9923$. ‡ $NH_3 = 1.0000$ for both because the target is constant — both models predict −40.0 trivially.*

C. Training Dynamics (SpectralResNet, 20K Run)

The model trained for the full 150-epoch budget without triggering early stopping. The training loss dropped from 1.594 at epoch 1 to 0.182 at epoch 150; validation loss fell from 0.350 to 0.175, indicating no significant overfitting. The 124K run showed dramatically faster convergence: training loss 45.76 at epoch 1 (unnormalised scale), dropping to 0.0119 at epoch 150 with a final validation loss of 0.000157.

Scale	Final Train Loss	Final Val Loss	RF mean R^2	ResNet mean R^2
20K (HPC)	0.182	0.175	0.9245	0.8226
124K (full)	0.0119	0.000157	0.9663	0.9923

Table VII. *Summary of training outcomes at both dataset scales.*

VI. DISCUSSION

A. H_2O Anomaly

H_2O is the most striking failure of the ResNet ($R^2 = -0.0008$ on the 20K run). The cause is statistical: the H_2O target has a standard deviation of only *0.0002 log_{10} units* across the dataset — a range of just 0.003 log units total. The entire variability is below the noise floor of the ResNet's Gaussian augmentation ($\sigma = 0.01$). The RF, which predicts the training-set mean, achieves apparent $R^2 = 0.7389$ simply because small deterministic corrections to the near-constant value reduce MSE slightly below the total variance. In practice, both models achieve RMSE ≈ 0.0001–0.0002 (sub-order-of-magnitude accuracy), but the near-zero denominator in $R^2 = 1 - MSE/Var$ makes the metric deceptive. This is a known limitation of R^2 as an evaluation metric for near-constant targets.

B. NH_3 Anomaly

NH_3 is at the log-floor (−40.0) for 100% of the 124,314 simulated samples — the ATMOS photochemistry model produces no detectable ammonia in this parameter space. The RF correctly learns to always predict −40.0, yielding $R^2 = 1.0$. The ResNet, using a probabilistic loss with dropout and weight noise, fails to converge to a trivial constant, resulting in $R^2 \approx 0$ and RMSE = 0.202 log units. This is a model design lesson: when a target is degenerate, a simple lookup model will always outperform a regularised deep model.

C. Biosignature Molecules

The biosignature molecules O_3 ($R^2 = 0.993$), CH_4 (0.9999), O_2 (0.975), and N_2O (0.988) are all accurately predicted by the ResNet at 20K scale. O_3 and CH_4 are particularly important — their co-detection (absent the other) is a false-positive biosignature, while co-detection of both simultaneously is among the strongest evidence for biogenic chemistry. The ResNet's near-perfect recovery of both suggests it has learned the underlying photochemical relationships that link these molecules to the atmospheric state encoded in the CLIMA profile.

D. Scaling Behaviour

The 20K → 124K improvement is dramatic: ResNet mean R^2 rises from 0.82 to 0.99, and RF from 0.92 to 0.97 (despite the RF still being trained on only 10,000 samples — the improvement in RF reflects the larger, more representative test set). This confirms the task is **data-limited at 20K**: the model has sufficient capacity but insufficient training diversity. Given the full INARA archive contains ~3.1 million samples, further scaling is a direct path to improved retrieval accuracy.

E. Computational Efficiency

Both models perform inference in **milliseconds per planet** once trained, compared to hours for Bayesian nested sampling. The ResNet processes a batch of 64 CLIMA profiles in ~2 ms on Apple Silicon MPS (single forward pass). The RF inference time is similarly fast (~1 ms for 12 parallel forest evaluations). This ~10,000× speedup over Bayesian methods makes population-level surveys (thousands of planets) computationally feasible.

F. Limitations and Future Work

- **Synthetic data only:** Models are trained on ATMOS simulations. Real JWST spectra differ in noise characteristics, instrument resolution, and systematic biases. Validation on JWST observations (e.g., WASP-39 b) is a critical next step.
- **No uncertainty quantification:** The current models produce point estimates only. Bayesian neural networks or MC Dropout could provide posterior distributions comparable to nested sampling.
- **PSG spectra:** A complementary dataset of 0.2–2.0 μm transmission spectra (4,378 wavelength points) is available from the same archive but not yet incorporated. Training on actual spectra would be more directly applicable to JWST observations.
- **H_2O and NH_3:** These targets require either dataset curation (filtering constant-target samples) or model redesign (dedicated constant-detection logic).

VII. CONCLUSION

We trained and evaluated two ML models for exoplanet atmospheric retrieval on the INARA ATMOS benchmark dataset. The 1D SpectralResNet and per-molecule Random Forest both achieve scientifically useful accuracy — $R^2 > 0.99$ on the full 124K dataset — at millisecond inference time, representing a ~10,000× speedup over Bayesian retrieval methods.

On the 20K HPC run, the ResNet outperforms the RF on 10/12 molecules when excluding the degenerate NH_3 target and near-constant H_2O. The mean R^2 advantage of the RF in the raw comparison (0.9245 vs. 0.8226) is entirely attributable to the RF's trivial handling of constant and near-constant targets, not superior modelling of atmospheric chemistry. At full 124K scale, the ResNet clearly dominates (mean R^2 0.9923 vs. 0.9663), particularly for low-variance molecules (N_2, H_2) where the deep model's learned representations capture fine-grained chemical signals that decision trees miss.

These results validate the INARA benchmark as a useful test bed for ML retrieval algorithms and demonstrate that data scale — not model capacity — is the primary bottleneck at this task. Future work should focus on validation against real JWST spectra, uncertainty quantification, and incorporation of the PSG spectral dataset for direct comparison with Bayesian retrieval posteriors.

REFERENCES

1. I. Zorzan et al., "INARA: Investigating Non-trivial Atmospheres for Retrieval Algorithms," *ApJS* 277:38, 2025.
2. N. Madhusudhan, "Exoplanetary Atmospheres: Key Insights, Challenges, and Prospects," *ARA&A* 57, 2019.
3. P. Márquez-Neila et al., "Supervised machine learning for analysing spectra of exoplanetary atmospheres," *Nature Astronomy* 2, 2018.
4. M. Vasist et al., "Neural posterior estimation for exoplanetary atmospheric retrieval," *A&A*, 2023.
5. T. D. Gebhard et al., "Flow matching for full posterior atmospheric retrieval," *A&A*, 2024.
6. JWST Transiting Exoplanet Community ERS Team, "Identification of carbon dioxide in an exoplanet atmosphere," *Nature* 614:649, 2023.